

Assignment 1: Replication exercise (25%)

Overview:

I want you to find a recent paper in development economics that is published in either the American Economic Review (not “Insights”) or one of the American Economic Journals (Applied, Policy, Macroeconomics, or Microeconomics), is no more than three years old by publication date, and has replication data and code on the journal website. I want you to tell me what the main question of the paper is (100 words), why this question is important (100 words), and what the paper's main empirical method is (100 words). Then, I want you to take five tables or figures from the paper and: tell me what the main point is of the table or figure (100 words), show that you can replicate the table or figure (100 words), and tell me what the main commands are used to make the table or figure (100 words). Then I want you to do a “broad” replication of the paper in a new context.

You will learn:

- How to teach yourself new techniques in that you saw in a paper.
- How to read a journal article, and understand its results.

Additional instructions:

- Do not use one of the required readings. I want you to read something new.
- I am not willing to sift through a large number of files for each student. I want this entire assignment to be submitted as a single pdf. So:
 - when you tell me "what the main point is of the table or figure," it would be good for you to include a screenshot of the table or figure next to your explanation.
 - when you "show that you can replicate the table or figure," I want to see the log file (or equivalent) that contains your output. But I don't want it as a separate file. I want it to be included as part of the text of the assignment, either in the main text or the appendix. You'll have to carefully format this so that it is legible. See how I've included log files in my lecture slides? That's what I want to see in your assignment. I've put all the Stata code for my lecture slides on the Moodle page, so you can see what you need to do to save nicely-formatted records of your output.
- Please use the following format for the title: "EC982: Replication of Dell, M., Lane, N., & Querubin, P. (2018). The historical state, local collective action, and economic development in Vietnam. *Econometrica*, 86(6), 2083-2121."
- Use your own words. We use software to check for plagiarism, and students have been caught and punished in the past in this module.
- I don't want the do file – that just shows me you can copy paste the code into a new document. I want the log file in which I can see the Stata output as well. This is how I

know you actually made the code run. If you are using a program other than Stata, the same principle applies: show me you actually ran the code successfully!

Assessment criteria:

There are 25 tasks I want you to complete, and you will receive one point for each of them:

1. The assignment is easy to read and understand, and does not contain any errors in spelling or grammar.
2. You have selected a paper for replication that fits the following criteria:
 - a. It is published in either the American Economic Review or one of the American Economic Journals.
 - b. It is no more than three years old.
 - c. It has replication data and code on the journal website.
 - d. It uses data from a developing country (which you can define as a country below the GDP per capita cutoff for classification as a “High-Income Economy” according to the World Bank).
3. The following formatting guidelines are followed (excepting for the log files, which simply need to be legible):
 - a. Times New Roman font.
 - b. 12 point font.
 - c. 1.15 spacing.
 - d. Fully justified (i.e. not ragged right).
 - e. One inch margins.
 - f. Page numbers are in the bottom centre of each page.
 - g. Do not include your name on the assignment.
 - h. Include your ID number on the assignment.

(Note: The document you are reading follows these guidelines).
4. In 100 words or fewer, you tell me the main question of the paper.
5. In 100 words or fewer, you tell me the paper’s main empirical method.
6. In 100 words or fewer, you tell me why the main question of the paper is important.

For the first figure or table that you choose to replicate:

7. In 100 words or fewer, you tell me the main point of the figure or table.
8. You include a log file showing you have successfully replicated the table or figure.
9. In 100 words or fewer, you tell me the main commands used.

For the second figure or table that you choose to replicate:

10. In 100 words or fewer, you tell me the main point of the figure or table.
11. You include a log file showing you have successfully replicated the table or figure.
12. In 100 words or fewer, you tell me the main commands used.

For the third figure or table that you choose to replicate:

13. In 100 words or fewer, you tell me the main point of the figure or table.

14. You include a log file showing you have successfully replicated the table or figure.
15. In 100 words or fewer, you tell me the main commands used.

For the fourth figure or table that you choose to replicate:

16. In 100 words or fewer, you tell me the main point of the figure or table.
17. You include a log file showing you have successfully replicated the table or figure.
18. In 100 words or fewer, you tell me the main commands used.

For the fifth figure or table that you choose to replicate:

19. In 100 words or fewer, you tell me the main point of the figure or table.
20. You include a log file showing you have successfully replicated the table or figure.
21. In 100 words or fewer, you tell me the main commands used.

The remaining points will be substantially more difficult to earn:

22. For one table or figure, demonstrate that you can replicate the result using a different program than the one used in the paper's replication files. For example, if the original paper was coded in Stata, use a program such as R or Matlab. This has no word limit. Please indicate in the main text of your assignment if you have done this: if it is only in the appendix, I may miss it.
23. Identify a context in which you can test the external validity of the main result of the paper. For example, if the paper uses a difference in difference approach to assess the effect of childhood exposure to a school-building program on later-life wages in Indonesia, you could use a similar program that occurred in Zambia and assess the impact on later-life wages there. Implement, as closely as possible, the principal method from the paper you are replicating in this alternative context. I do not mean that I want you to find a second study to replicate. I mean that I want you to find original data. This has no word limit.
24. For this same alternative context, implement as closely as possible the analyses from all tables and figures in the paper you are replicating. This has no word limit.
25. The results of this broad replication are sufficiently informative that, expanded to a full-length paper, the results could be published in a journal as well ranked as the *Journal of Development Economics*.

Note: If a point has several parts (e.g. point 3 above), you must complete all parts to receive a point, and I will not give partial credit. If you put page numbers in the top right, you lose the whole point. Word limits are strict; if you complete point 15 in 102 words, and I notice, you lose the point.

Also note: I expect almost everyone to do this in Stata. But if your paper uses a different program (e.g. R, Matlab, Gauss, Python, Assembly...) for its analysis, that's fine. Unfortunately, I have much more knowledge of Stata than any of the others so I may not be helpful with the others.

Obviously, additional penalties may be applied for plagiarism, similarity to other texts, and other poor academic practices.

Assignment 2: Required Reading (25%)

Overview:

This portion of the coursework exists in order to ensure that you have carefully read through the required readings. The readings are listed on Moodle, change every year, and are also discussed in my slides for the current year. I have prepared twenty review questions, one for each reading.

You will learn:

- “General economics knowledge” relating to the topics of the review articles.
- How to recognize good science and evaluate factual and causal claims.

Additional instructions:

- To earn most of the points below, you must provide an “exceptional” answer. I will grade this assignment strictly, and I will not give partial credit except as otherwise indicated.
 - An exceptional answer either:
 - gives me the “correct” answer I have in mind, with all of its key parts, or
 - gives me an answer that I think is better than the “correct” answer I had in mind.
 - For example, in a previous year I asked the following about Martinez (2022, JPE): “How are these results relevant to the literature on foreign aid?”. The key points of the “correct” answer I had in mind were:
 - “A good answer would discuss Table 4, which shows (in a triple interaction) that countries increasingly overstate GDP growth after they cross the IDA eligibility threshold. Below the threshold, overstating GDP could make them ineligible for aid. Indeed, the greater NTL gradient for autocracies only emerges after crossing the threshold. A good answer would also discuss Table 5, which shows a null effect of crossing the IDA eligibility threshold, and thus losing aid, on growth. But this null is only apparent once corrected GDP data that account for manipulation are used.”

An answer that only discussed Table 4 and not Table 5 did not earn the point.
- A “good-faith” answer, by contrast, is a real attempt to answer the question that shows you have read the reading, but that is not necessarily exceptional. It is an answer that showed you at least tried in good faith.
- I won’t grade you on formatting for this assignment. I don’t care about fonts, margins, etc... That said, your document will be much nicer to look at and easier to read with a large serif font, moderate spacing, and block justification.

- And a long answer that mentions several points, in the hope that some of them are correct, is unlikely to be exceptional. Do not just throw everything at the wall and hope something sticks.
- I want a single pdf with your student number on it.

Assessment criteria:

- Point 1: Provide an exceptional answer, based on Ghatak and Mookherjee (2025), to the question “How do connections between firms moderate the impacts of government interventions?”
- Point 2: Provide an exceptional answer, based on Kala (2025), to the question “How do managers that achieve autonomy raise more capital?”
- Point 3: Provide an exceptional answer, based on Breza and Kaur (2025), to the question “Are labour markets in developing countries useful for self-insurance?”
- Point 4: Provide an exceptional answer, based on Rexer (2025), to the question “What are the net benefits of indigenization?”
- Point 5: Provide an exceptional answer, based on Banerjee et al (2024), to the question “What is missing from Banerjee et al’s (2024) theoretical analysis of redistribution?”
- Point 6: Provide an exceptional answer, based on Burlig and Preonas (2024), to the question “What are the returns to rural electrification?”
- Point 7: Provide an exceptional answer, based on Besley et al. (2022), to the question “Why are connections between enterprises, bureaucrats, and civil society important in developing countries?”
- Point 8: Provide an exceptional answer, based on Deserranno et al. (2025), to the question “How do we know that managers make workers more productive?”
- Point 9: Provide an exceptional answer, based on Hjort and Tian (2025), to the question “How does the internet affect state capacity?”
- Point 10: Provide an exceptional answer, based on Gertler et al (2024), to the question “Would the presence of multiple lenders change the impacts of digital collateral?”
- Point 11: Provide an exceptional answer, based on Rohner (2024), to the question “How can governments use the evidence from the literature on outside interventions?”
- Point 12: Provide an exceptional answer, based on Blattman et al (2025), to the question “What are the strengths and weaknesses of the data in this paper?”
- Point 13: Provide an exceptional answer, based on Jayachandran (2022), to the question “How do tariff and non-tariff barriers affect the environment?”
- Point 14: Provide an exceptional answer, based on Greenstone et al. (2025), to the question “How do the authors measure the empirical relationship between the cost of an incremental reduction in emissions and the total quantity of emissions?”
- Point 15: Provide an exceptional answer, based on Desmet and Rossi-Hansberg (2024), to the question “How do spatial considerations influence how governments should respond to climate change?”
- Point 16: Provide an exceptional answer, based on Annan (2025), to the question “Does anti-misconduct information help households deal with unanticipated adverse events?”

- Point 17: Provide an exceptional answer, based on Chen and Zha (forthcoming), to the question “How have the sources of misallocation in China’s economy changed over time?”
- Point 18: Provide an exceptional answer, based on Kaur et al. (2025), to the question “Why is this the right context in which to answer the research question?”
- Point 19: Provide an exceptional answer, based on Galor (2025), to the question “How can Unified Growth Theory explain differences in comparative development?”
- Point 20: Provide an exceptional answer, based on Khan (2025), to the question “Why should we think the effect of treatment works through preferences, and not some other channel?”
- Points 21-25: For every reading for which you provide an answer that is either “good faith” or “exceptional,” you will earn 0.25 points. So: if you give every question an honest try, you can get no less than 20% on the assignment.

Obviously, additional penalties may be applied for plagiarism, similarity to other texts, and other poor academic practices.

Assignment 3: Essay (25%)

Overview:

I want you to find a question of the form "What is the effect of X on Y?" that is relevant to development economics. For example, "What is the effect of crop insurance on agricultural investment?"

I want you to summarize:

- What the question is, why it important, and what the key lessons are from the literature reviewed (100 words).
- The results obtained by studies that use randomized controlled trials (450 words).
- The results obtained by studies that use natural experiments (450 words).
- The results obtained by other studies (450 words).

Then, I want you to discuss whether these differ and why/why not (450 words).

You will learn:

- What the state of the literature is on a topic that interests you, and that I might not cover in the lectures.
- Why methods matter, and why we should trust some results more than others.

Additional instructions:

You cannot work alone on this assignment. You can work in groups of 2 or 3, but I want no group to be comprised entirely of students from the same country. Why?

- I want you to be able to form a study group as a result of this assignment. This group will help you on your other assignments. This group will also hopefully help you in other modules, and on your thesis.
- If not all students have the same native language, I believe the writing and grammar of the essay will improve.
- Your undergraduate institutions all had their own strengths and weaknesses. These often differ by country. A mixed group will help you share your strengths.
- I want you to form networks and develop soft skills as a result of this degree. These will help you in your career. Self-segregating by country will make it harder to gain soft skills.

It is OK for 2 members of the group to be from the same country so long as the third is not.

If you find yourself unable to form a group that is not all students from the same country, I want you to form an arrangement with another group that you will proofread and provide

feedback on each other's essays before you submit, and provide me with evidence by email that you have done this.

When you submit, one student should submit the essay and make sure it has all three student IDs on it. The other students should simply submit a page stating the IDs of their group members. For example: "I am in a group with 1573813 and 3183751."

- I don't care what format you use for your bibliography: just use one that is consistent and easy to understand.
- While you are free to draw on literature from outside economics, your review should focus on contributions published in core economics journals such as the Quarterly Journal of Economics, Econometrica, the Journal of Political Economy, the American Economic Review, the Review of Economic Studies, the various American Economic Journals, the Economic Journal, the Review of Economics and Statistics, and the Journal of the European Economic Association.
- Since the essay should reflect the state of knowledge, be sure to cite recent work and not solely older, foundational papers.
- You will also find that some journals frequently publish reviews of the literature (e.g. the Annual Review of Economics, the Journal of Economic Literature, and the Journal of Economic Perspectives). These, and Google scholar, will be good places to start in identifying the important papers to read and the more recent work that has followed from older papers.
- Although I say that I "want you to summarize the results obtained by..." I don't want to read a series of unthinking, short abstracts. Synthesize as you go.
- Use your own words. We use software to check for plagiarism, and students have been caught and punished in the past in this module.

Assessment criteria:

While the university does have an official scale for essay-based work, I believe this scale allows me to be more inconsistent across essays than I would like. Grading an essay will always be an unavoidably subjective task, but I can constrain my subjectivity. So, I will assess the essay by enumerating 25 tasks I want you to complete, and telling that you will receive one point for each of them. I have designed this scale to target a lower than usual distribution for an essay assignment. Choose this target because I expect some other assignments (e.g. the replication assignment) to have higher means.

1. The essay obeys the following formatting rules: Times New Roman, 12pt, 1.15 spacing, fully justified (i.e. not ragged right), with one inch margins and page numbers. Names do not appear the assignment or in the filename. ID numbers are included on the assignment. The document you are reading follows these guidelines. Note: if you fail to adhere to any one of these rules, you will lose the whole point.

2. The essay is focused on a central question of the form "What is the effect of X on Y?" that is relevant to development economics. For example, "What is the effect of crop insurance on agricultural investment?"
3. In the central question of the essay, X and Y are defined narrowly enough that it is meaningful to compare the studies reviewed.
4. The first section of the essay is an introduction of no more than 100 words that states what the question is and why it is important.
5. The first section also summarizes the key lessons are from the literature reviewed.
6. The second section of the essay is no more than 450 words long and summarizes critically the results from studies that address the central question using randomized controlled trials.
7. The third section of the essay is no more than 450 words long and summarizes critically the results from studies that address the central question using natural experiments.

While the concept of a natural experiment is somewhat fuzzy, I have in mind studies in which manipulation of treatment was not conducted by the researcher, but in which the claims to exogeneity and causal inference are as good as, or almost as good as, those from an RCT. Examples include assignment lotteries carried out by policymakers, weather shocks in panel data, and convincing regression discontinuities.

8. Sources of exogenous variation used by the "natural experiment" papers are clear to the reader in this section.
9. The fourth section of the essay is no more than 450 words long and summarizes critically the results from studies that address the central question using other approaches. These can include theoretical papers, anthropology, descriptive "correlates of" papers, qualitative work, or any other work where claims to causal inference are not as convincing as in an RCT or natural experiment.
10. The fifth section of the essay is no more than 450 words and is a "discussion" section that explains whether the results obtained by the three broad types of study (RCT, natural experiment, other) differ and, if so, why.
11. The bibliography includes at least 15 papers, and least half the works cited are:
 - a. in leading research journals such as Quarterly Journal of Economics, Econometrica, the Journal of Political Economy, the American Economic Review, the Review of Economic Studies, the various American Economic Journals, the Economic Journal, the Review of Economics and Statistics, and the Journal of the European Economic Association, or...
 - b. in other respectable mainstream economics journals like the Journal of Development Economics, World Development, or and Economic Development and Cultural Change, or in leading literature review journals such as the Annual Review of Economics, the Journal of Economic Literature, and the Journal of Economic Perspectives.

This is not an exhaustive list; it is only meant to be indicative. While there is no one accepted guide to which economics journals are "good", these links can get you started:

<https://ideas.repec.org/top/top.journals.hindex.html>

12. At least half the works cited have been published within the last 10 years.
13. At least half the works cited use data from developing countries (which you can define as countries that are below the GDP per capita cutoff for classification as a “High-Income Economy” according to the World Bank).
14. The essay demonstrates analytical insights throughout, noting strengths, limitations, and exceptional characteristics of the studies as they are discussed, i.e. in sections two through five, not only in the “discussion” section.
15. The essay does not read as a sequence of summaries, but instead synthesizes the papers cited as it goes along.

An example of an essay that did this well in the past is one that considered the impact of microfinance on household welfare. Each subsection was divided into three paragraphs, one considering the effects each study found on enterprise creation, one considering the effects each study found on income, and one considering the results each study found on consumption.

16. The points made in the discussion section demonstrate understanding of the strengths and weaknesses of RCTs and other approaches, such as those discussed in Glennerster and Takavarasha.
17. The points made in the discussion section are not only stated in abstract terms, but rather highlight how differences between specific studies cited can be understood.
18. The “discussion” section provides a quantitative analysis of the reasons why different papers studied have come to different conclusions.

An example of an essay that did this well in the past is one that considered the effect of conditional cash transfers on school enrolment. The essay included a scatterplot in which effect sizes and confidence intervals from the various studies were plotted on the y axis, the initial level of enrolment was plotted on the x axis, and a line of best fit was shown, demonstrating smaller effects in areas where school enrolment was already at a high level.

19. The points made in the “discussion” section show insight that sets it apart from other essays that have been submitted in the MSc programme over the past several years. To earn this point, the essay must make claims that are not obvious to the grader, or that require concepts and reasoning that go beyond what is taught in EC982.
20. There are very few errors in spelling, grammar, or typography.
21. The essay stands out for being clearly and engagingly written, easy to follow, and effective at communicating the author’s ideas. Writing of this quality regularly appears in publications such as The New Yorker or The Atlantic.
22. The grader believes that, taken as a whole, this is an essay of rare, exceptional, outstanding quality.

23. The essay is sufficiently novel, important, well-written, and well-argued that it would reasonably be expected by any reader to have a real world impact, influencing the decisions of policymakers or practitioners.
24. The essay is sufficiently novel, important, well-written, and well-argued that it would reasonably be expected by any reader to receive wide acclaim in the popular press or the economics blogosphere.
25. The essay is sufficiently novel, important, well-written, and well-argued that it would reasonably be expected by any reader to be solicited for publication in a leading journal that publishes literature reviews in economics, such as the Journal of Economic Literature or the Annual Review of Economics.

Note: If a point has several parts, you must complete all parts to receive a point, and I will not give partial credit. Word limits are strict; if you complete point 6 in 451 words, and I notice, you lose the point.

Obviously, additional penalties may be applied for plagiarism, similarity to other texts, and other poor academic practices.

Assignment 4: Empirical Assignment (25%)

Overview:

In this assignment, you will look at correlations – some plausibly causal – related to a topic of importance to development economics: conflict in Africa. This assignment draws on the methods I teach in the “How to” sections of the module, but also forces you to learn some new skills that I do not cover. This assignment consists of 25 questions. The answer to each of these questions is a number.

You will learn:

- New techniques, such as how to work with spatial data.
- How to independently improve your empirical skills when faced with a new task, without guidance or direction.
- How to do well on the coding test for a JPAL-type interview.

Additional instructions:

Your submission will consist of two files. The first file will be a pdf. It will contain a log (or similar) file demonstrating that you were able to produce these answers yourself in Stata (or a similar program, or a mix of programs). The second file will be a csv. The name of the file will be your student number. The first column of the csv will be the question number, from 1 to 25. The second column will be your numeric answer to the question. This number must be formatted as a number without alphabetic characters. So: no explanatory notes. No scientific notation. The title of the first column should be “q” and the title of the second column should be “a”.

For this assignment, I have put several datasets online:

- borders_tribes: a map of ethnic groups in Africa, from Murdock (1959).
- GEDEvent_v25_1: a set of conflict events from Davies et al. (2025) and Sundberg and Melander (2013).
- world-administrative-boundaries: A map of country borders.
- Extended_Atlas: Walther’s (2005) extended version of the ethnographic atlas.
- tri: Ruggedness data from Nunn and Puga (2012).
- chirps: A folder of rainfall data from CHIRPS.
- HCED Master Data: Miller and Bakar’s (2022) historical conflict data.
- HMIsea: travel time walking or by traditional seafaring from Özak (2018, 2010).

The main papers inspiring this assignment are:

- Besley, T., & Reynal-Querol, M. (2014). The legacy of historical conflict: Evidence from Africa. *American Political Science Review*, 108(2), 319-336.

- Depetris-Chauvin, E. (2015). State history and contemporary conflict: Evidence from sub-saharan Africa. Available at SSRN 2679594.
- Harari, M., & Ferrara, E. L. (2018). Conflict, climate, and cells: a disaggregated analysis. *Review of Economics and Statistics*, 100(4), 594-608.
- Michalopoulos, S., & Papaioannou, E. (2016). The long-run effects of the scramble for Africa. *American Economic Review*, 106(7), 1802-1848.
- Nunn, N., & Puga, D. (2012). Ruggedness: The blessing of bad geography in Africa. *Review of Economics and Statistics*, 94(1), 20-36.

Start by creating a grid of latitude-longitude points at unit intervals. Latitude should range from -50 to 40, and longitude should range from -20 to 65. From this set of points, extract the subset that intersects with `borders_tribes`. That is: find the set of points that is over land in Africa, according to the map.

- *Question 1: How many points intersect borders_tribes?*

Create a set of grid cells, all 1 degree in latitude by 1 degree in longitude, whose centroids are this subset of African points. Clip the cells so that they do not extend outside of `borders_tribes`.

- *Question 2: What is the average area across cells?*

For each grid cell in this set of clipped cells, compute the probability that any two points selected at random are in the homelands of different ethnic groups. Call this probability *HHI*. Across grid cells, compute the average of this probability.

- *Question 3: What is the average, across grid cells, of the probability that two points, selected at random, are in the homelands of different ethnic groups?*

Using the file `GEDEvent_v25_1`, compute the number of civilian deaths in each grid cell in each year from 1989 to 2024, inclusive.

- *Question 4: What is the standard deviation of civilian deaths, across grid cells, in 1999?*

Sum all civilian deaths in a grid cell across years. Compute the inverse hyperbolic sine of the number of total civilian deaths for the grid cell. Call this variable *IHS(Deaths)* from now on. Regress this cross-sectional measure of civilian violence on *HHI*, which you computed earlier.

- *Question 5: What is the coefficient on HHI?*

Alternatively, estimate a Poisson model in which the outcome is the number of total civilian deaths for the grid cell, and *HHI* is the independent variable.

- *Question 6: What is the coefficient on HHI in this alternative specification?*

Those look surprisingly different! Try winsorizing the number of total civilian deaths for the grid cell at the 99th percentile. Estimate a Poisson model in which the outcome is this winsorized variable, and *HHI* is the independent variable.

- *Question 7: What is the coefficient on HHI in this third specification?*

Using the “name” variable in world-administrative-boundaries, find the country into which each grid cell’s defining latitude-longitude point (i.e. the integer values of latitude and longitude from Question 1) falls. Run a regression where the dependent variable is *IHS(Deaths)*. The independent variable is *HHI*. Control for country fixed effects.

- *Question 8: What is the coefficient on HHI in this specification with country fixed effects?*

Run the same regression of *IHS(Deaths)* on *HHI* and country fixed effects as in Question 7, but use Conley (1999) standard errors to allow for spatial dependence. In particular, use a Bartlett kernel with radius 500 km over great-circle distances.

- *Question 9: What is the standard error of the coefficient on HHI?*

Using the Extended_Atlas file, identify all ethnic groups whose coordinates intersect those of borders_tribes. Variable EA033 in this dataset is a measure of jurisdictional hierarchy – a proxy for state capacity. Treat the unique combination of ethnic group name, latitude, and longitude as a separate observation. For example, treat the Soninke at (12, -14.5) as distinct from the Soninke at (12, -1.5). For combinations of name, latitude, and longitude that do not uniquely identify observations, drop observations that are missing the jurisdictional hierarchy variable. For combinations of name, latitude, and longitude that do not uniquely identify observations, keep only the observation with the maximum value of the jurisdictional hierarchy variable.

- *Question 10: What is the mean value of jurisdictional hierarchy across ethnic groups in this de-duplicated African sample?*

Build Thiessen polygons around these ethnic groups, clip these polygons using borders_tribes, and compute the area of the clipped polygons in square kilometres.

- *Question 11: What is the mean area of an ethnic group polygon in this sample?*

For each Thiessen polygon, compute the total number of civilian deaths that occur across all years according to GEDEvent_v25_1. Take the inverse hyperbolic sine of that total. Abusing notation, call this variable *IHS(Deaths)*. Run a regression where the outcome variable is

$IHS(Deaths)$, the independent variable is jurisdictional hierarchy, and again use Conley (1999) standard errors with a Bartlett kernel and a 500 km cutoff.

- *Question 12: What is the standard error of the coefficient on jurisdictional hierarchy?*

Now use `tri` to compute the mean ruggedness for each of the Thiessen polygons. Call this mean at the polygon level *Ruggedness*.

- *Question 13: What is the mean value of Ruggedness across polygons?*

Run the same regression as above ($IHS(Deaths)$ on jurisdictional hierarchy, and Conley standard errors with a 500km cutoff), but add ruggedness as a control variable.

- *Question 14: What is the standard error of the coefficient on jurisdictional hierarchy in this regression that controls for ruggedness?*

Intersect the ethnic groups from `borders_tribes` with the country boundaries from `world-administrative-boundaries`. Compute the area of every ethnicity-country pair in this intersection. Identify the “main” country for any ethnic group as the one in which the largest percentage of its area is located. Compute the share of each ethnicity’s area that lies in each country. Identify all ethnic groups for whom at least 10% of the group’s area is in a country other than its main country. Code a variable, *PARTITIONED*, that is equal to 1 if the ethnic group is thus split, and zero otherwise.

- *Question 15: Among partitioned ethnic groups, what is the mean share of the group’s area that is in its “main” country?*

For each polygon, compute the total number of civilian deaths that occur across all years according to `GEDEvent_v25_1`. Take the inverse hyperbolic sine of that total. Again abusing notation, call this variable $IHS(Deaths)$.

Run a regression at the polygon level of $IHS(Deaths)$ on *PARTITIONED* and on country fixed effects. Cluster standard errors by ethnicity.

- *Question 16: What is the standard error of the coefficient on the “partitioned” dummy?*

Go back to the data you created on civilian deaths by grid cell, between 1989 and 2024. Using the data in the “chirps” subfolder, compute the value of rainfall for each grid cell in each year by averaging over raster points within a cell.

- *Question 17: What is the standard deviation of rainfall across grid cells in 1999?*

Take the inverse hyperbolic sine of civilian deaths in each year for each grid cell. Again abusing notation, call this variable $IHS(Deaths)$. Run a regression of $IHS(Deaths)$ on rainfall, on grid cell fixed effects, and on year fixed effects. Cluster standard errors by grid cell.

- *Question 18: What is the standard error of the coefficient on rainfall?*

Take the variable HHI , which you computed earlier. Run a regression of $IHS(Deaths)$ on rainfall, on the interaction of rainfall and HHI , on grid cell fixed effects, and on year fixed effects. Cluster standard errors by grid cell.

- *Question 19: What is the standard error of the coefficient on the interaction of rainfall and HHI ?*

Identify every battle in HCED Master Data that intersects with borders_tribes. For each grid cell that you constructed earlier, compute the physical distance in kilometres from its defining latitude-longitude point (i.e. the integer values of latitude and longitude from Question 1) to every battle. Then measure the travel time from these defining latitude-longitude points to the every battle.

- *Question 20: What is the mean travel time across cell-battle pairs that are within 500 km of each other?*

For each cell, compute an inverse travel time measure of the cell's exposure to the battle as one divided by the travel time. For each grid cell, sum over this inverse distance measure across battles to compute a measure of historical conflict exposure for each cell.

In the cross-section of grid cells, run a regression of $IHS(Deaths)$ on historical conflict exposure and country fixed effects as defined for question 8. Again, use Conley (1999) standard errors, i.e. a Bartlett kernel with radius 500 km over great-circle distances.

- *Question 21: What is the standard error on historical conflict exposure?*

Following instructions on <https://github.com/Seshat-Global-History-Databank/cliopatria>, open the file cliopatria_polities_only.geojson. Note that I have not provided you with this file. Use the variable "Name" to identify distinct polities.

- *Question 22: How many distinct polities are in these data?*

Identify the polities in cliopatria_polities_only.geojson that intersect the clipped grid cells created in Question 1.

- *Question 23: How many distinct polities in the Cliopatria data intersect the clipped grid cells?*

For every year between 1500 and 1884, inclusive, identify the set of polities that intersect the grid cell. For each grid cell, compute *Instability* as the share of years in which the set of polities differs from the previous year. This variable is a measure of historical political instability.

- *Question 24: What is the mean level of Instability across grid cells?*

In the cross-section of grid cells, run a regression of *IHS(Deaths)* on *Instability* and country fixed effects as defined for question 8. Again, use Conley (1999) standard errors, i.e. a Bartlett kernel with radius 500 km over great-circle distances.

- *Question 25: What is the standard error on Instability in this regression?*

Assessment criteria:

The assignment is graded out of 25. You will get a maximum of 1 point per question. Your grade will depend on how close your answer is to the correct answer in terms of absolute percent deviation. In particular, for each question, if your answer is $A \neq 0$ and the correct answer is C , your grade will be:

$$\frac{1}{1 + \theta \left| \frac{A - C}{C} \right|}$$

If your answer is $A = 0$, or if it is not a real number, your grade will be zero. θ is a tuning parameter that I will use to avoid being too harsh.

Obviously, additional penalties may be applied for plagiarism, similarity to other texts, and other poor academic practices. In particular, if you do not submit code showing that you were able to produce the answer you have submitted, you will not receive any points.